SINGLE-PHASE HYBRID ESS

STACK20

6 kW Hybrid Inverter + 20.48 kWh $LiFePO_4$ Battery System

A complete, pre-integrated energy storage system: a 6 kW single-phase hybrid inverter, four 5.12 kWh Talis 5 LFP battery modules and DC / PV protection, housed in one lockable cabinet. Built for solar self-consumption and whole-premises backup in homes and small businesses.



6 kW Hybrid | 20.48 kWh LFP | 2 MPPT · 9.6 kW PV | Wi-Fi · RS485 · CAN

6 kW	20.48 kWh	4 × Talis 5	51.2 V
HYBRID INVERTER	NOMINAL BATTERY ENERGY	5.12 KWH LFP MODULES	400 AH BATTERY BANK

PRODUCT HIGHLIGHTS

Integrated design
Hybrid inverter, four battery modules and DC / PV protection in a single, pre-wired cabinet.

20 kWh-class storage
Four 51.2 V 100 Ah Talis 5 LFP modules in parallel — 20.48 kWh nominal energy.

Smart battery management
Individual BMS in every module with CAN and RS485 / Modbus communication for inverter link and parallel operation.

Hybrid & backup operation
PV self-consumption, grid-connected and backup / off-grid modes with 2× surge for 10 s.

Service-friendly
Front-access battery modules and a dedicated electrical and protection section.

STACK platform
Clean, organised cabinet architecture for a neat, repeatable ESS installation.

MADE FOR HOMES & SMALL BUSINESSES

Homes & villas | Shops & showrooms | Small offices | Godowns & warehouses | Clinics & salons

Technical Specifications

STACK20 · 6 kW hybrid inverter + 20.48 kWh LFP battery system

SYSTEM

System type	Single-phase hybrid ESS
Rated inverter output	6 kW
Battery nominal energy	20.48 kWh
Battery bank	51.2 V, 400 Ah
Battery chemistry	$LiFePO_4$ (LFP)
Architecture	4 modules in parallel + hybrid inverter
Operating modes	Self-consumption, grid-tied, backup / off-grid†
Applications	Residential & small commercial

HYBRID INVERTER · 6 KW

Rated AC input / output power	6,000 W
Max. AC apparent power	6,600 VA
Rated AC current	26.1 A
Max. AC current	28.7 A
Max. continuous AC pass-through	40 A
Off-grid peak power	2 × rated, 10 s
AC nominal voltage / frequency	230 V / 50 Hz
THDi	< 3% of nominal power
Battery voltage range	40 – 60 V
Max. charge / discharge current	100 A
Max. PV input power	9,600 W
Max. PV access power	12,000 W
Max. PV input voltage	500 V
MPPT voltage range	150 – 425 V
MPPT trackers / strings	2 / 1 + 1
Max. PV operating current	30 A / 28.7 A
Max. efficiency	97.6%
MPPT efficiency	> 99%
Communication	Wi-Fi, RS485, CAN
Display	LCD
Ingress protection	IP65
Operating temperature	-40 to +60 °C (derating > 45 °C)
Cooling	Natural convection

BATTERY · TALIS 5 MODULES

Module	Talis 5 · 51.2 V / 100 Ah / 5.12 kWh
Cell configuration	3.2 V 100 Ah LFP · 16S1P per module
Number of modules	4, in parallel
System capacity	400 Ah @ 51.2 V
System nominal energy	20.48 kWh
Usable energy	≈ 16.4 kWh @ 80% DoD
Charge cut-off voltage	58.4 V per module
Discharge cut-off voltage	44 V per module
Max. charge current	Up to 100 A per module (settable)
Continuous discharge current	Up to 100 A per module (settable)
Cycle life	≥ 2,500 cycles @ 80% DoD*
Communication	CAN, RS485 / Modbus
Module dimensions	480 × 482 × 137 mm
Module weight	≈ 38.5 kg
Installation	Rack-mounted inside cabinet

PROTECTION

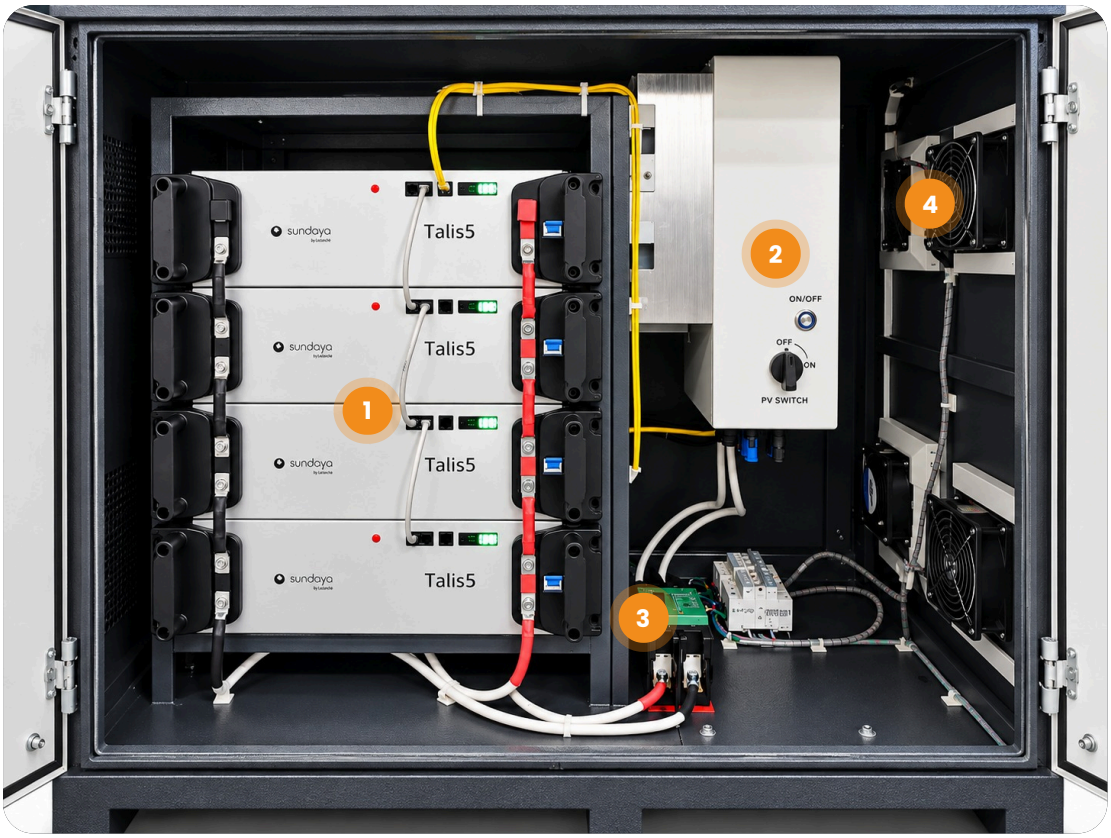
Battery (BMS)	Over / under-voltage, over-charge and over-discharge current, charge-current limiting, short circuit, overload
Inverter	DC reverse polarity, AC over-current, AC over-voltage, over-temperature, short circuit, DC-component monitoring, anti-islanding, residual-current detection†

COMMUNICATION

Battery	CAN / RS485 (Modbus)
Inverter	Wi-Fi / RS485 / CAN

Inside STACK20

Pre-integrated battery, inverter and protection sections



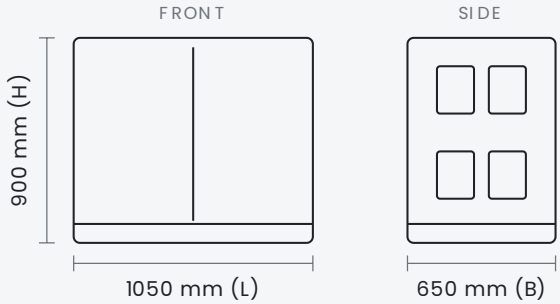
1 Battery section
4 × Talis 5 LFP modules, 51.2 V 100 Ah each, rack-mounted with front access

2 Hybrid inverter
6 kW single-phase hybrid inverter with ON/OFF and PV isolator switch

3 DC / PV protection
Dedicated protection and wiring section with DC busbars

4 Ventilation
Side louvres and fans for cabinet air circulation

CABINET DIMENSIONS



MECHANICAL

Cabinet (L × B × H)	1050 × 650 × 900 mm
Battery modules	4 × 480 × 482 × 137 mm
Battery weight	≈ 154 kg (4 × 38.5 kg)
Battery access	Front access, rack-mounted
Inverter ingress rating	IP65

NOTES

- Nominal energy = nominal voltage × rated capacity. Usable energy shown at 80% DoD.
- Battery-side power is limited by the inverter's 100 A battery current (≈ 5.1 kW at 51.2 V nominal); PV can supply the balance of the 6 kW output.
- * Cycle life per Talis 5 module specification at 80% DoD; subject to the Touch Energy commercial warranty terms.
- † Off-grid, anti-islanding and residual-current functions depend on inverter configuration and local grid approval.
- Install indoors or in a sheltered location; battery modules are rated for indoor use.
- Touch Energy reserves the right to change specifications without prior notice.